Well-ordering induction: a well-order satisfies the transfinite induction schema

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment propwoinduction (source lines 1265-1271); the proof environment begins two lines below it and is its own proof.

Editorial note. The proof prints its closing \end{proof} on the same line as its last sentence (source line 1277), so the retained proof span runs to line 1282 and thus keeps the author's following paragraph, which comments on the result, together with the proof; that paragraph contains no further assertion and is retained only so the extraction is not cut mid-paragraph.

Editorial note. The statement and the proof contain no \olref and no \cite, so no source pointer and no bibliography entry are needed: every symbol used (\lif, \lnot, \Setabs, !!{element}) is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit: OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have. Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's set-theory/ordinals unit, counted once: over any well-order, the transfinite induction schema holds.

Allowed standard prerequisites (editorial).

- Well-order. A relation $<$ well-orders A when it is a strict linear order on A in which every non-empty subset of A has a $<$-least element.. The statement assumes $<$ well-orders A and the proof uses the existence of a $<$-minimal element of the non-empty set X of counterexamples (source line 1276).
- Transfinite induction schema over a well-order. The schema asserted here: if for every a in A, (for all b $<$ a) phi(b) implies phi(a), then phi(a) holds for every a in A, for an arbitrary formula phi with parameters.. This is the counted proposition itself; the proof establishes it by contraposition rather than by a direct induction, so no further induction principle is assumed.

- Set abstraction and non-empty subset. X = {x in A : not phi(x)} is the subset of A collecting the counterexamples to phi, and phi may carry parameters.. The proof forms this set at source lines 1275-1276 and applies well-ordering to it once it is known to be non-empty.

Proposition 0.1. *If $<$ well-orders A, then for any formula $\phi(x)$:*

$$\textit{if } (\forall a \in A)((\forall b < a)\phi(b) \to \phi(a)), \textit{ then } (\forall a \in A)\phi(a).$$

Proof. We will prove the contrapositive. Suppose $\neg(\forall a \in A)\phi(a)$, i.e., that $X = \{\, x \in A \mid \neg\phi(x) \,\} \neq \emptyset$. Then X has an $<$-minimal !!element, a. So $(\forall b < a)\phi(b)$ but $\neg\phi(a)$. □

This last property should remind you of the principle of strong induction on the naturals, i.e.: if $(\forall n \in \omega)((\forall m < n)\phi(m) \to \phi(n))$, then $(\forall n \in \omega)\phi(n)$. And this property makes well-ordering into a very *robust* notion.[1]

[1] A reminder: all formulas can have parameters (unless explicitly stated otherwise).